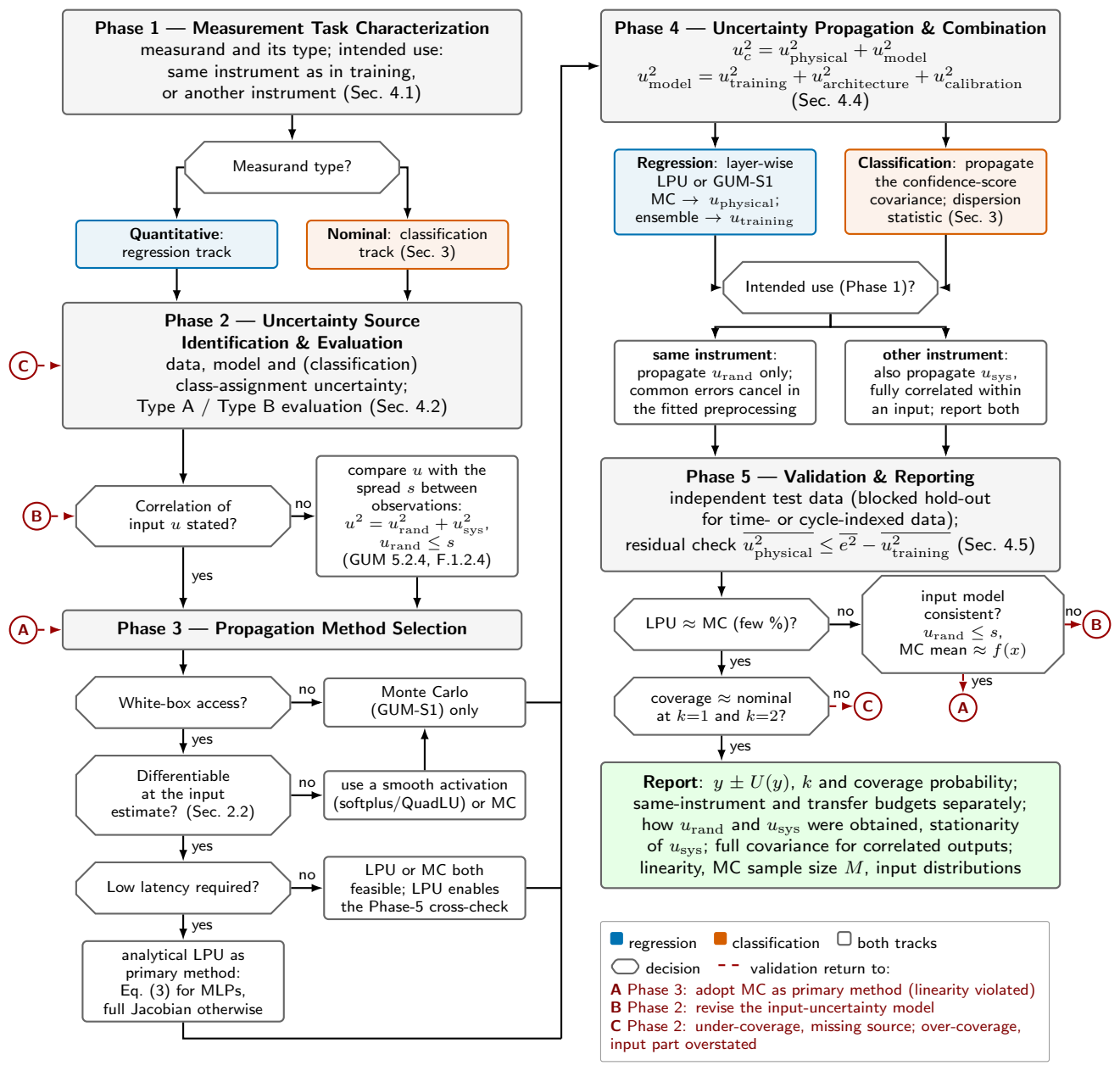

Phase 1 — Measurement Task Characterization
measurand and its type; intended use:
same instrument as in training,
or another instrument (Sec. 4.1)
Measurand type?
Quantitative: regression track
Nominal: classification track (Sec. 3)
Phase 2 — Uncertainty Source Identification & Evaluation
data, model and (classification) class-assignment uncertainty;
Type A / Type B evaluation (Sec. 4.2)
C
Correlation of input u stated?
B
no
compare u with the spread s between observations:
$u^2 = u^2_{\text{rand}} + u^2_{\text{sys}}$, $u_{\text{rand}} \le s$
(GUM 5.2.4, F.1.2.4)
yes
A
Phase 3 — Propagation Method Selection
White-box access?
no
Monte Carlo (GUM-S1) only
yes
Differentiable at the input estimate? (Sec. 2.2)
no
use a smooth activation (softplus/QuadLU) or MC
yes
Low latency required?
no
LPU or MC both feasible; LPU enables the Phase-5 cross-check
yes
analytical LPU as primary method:
Eq. (3) for MLPs,
full Jacobian otherwise
Phase 4 — Uncertainty Propagation & Combination
$u_c^2 = u^2_{\text{physical}} + u^2_{\text{model}}$
$u^2_{\text{model}} = u^2_{\text{training}} + u^2_{\text{architecture}} + u^2_{\text{calibration}}$
(Sec. 4.4)
Regression: layer-wise LPU or GUM-S1
MC → u_{physical}; ensemble → u_{training}
Classification: propagate the confidence-score covariance; dispersion statistic (Sec. 3)
Intended use (Phase 1)?
same instrument: propagate u_{rand} only; common errors cancel in the fitted preprocessing
other instrument: also propagate u_{sys}, fully correlated within an input; report both
Phase 5 — Validation & Reporting
independent test data (blocked hold-out for time- or cycle-indexed data);
residual check $\overline{u^2_{\text{physical}}} \le \overline{e^2} - \overline{u^2_{\text{training}}}$ (Sec. 4.5)
LPU ≈ MC (few %)?
no
input model consistent?
$u_{\text{rand}} \le s$, MC mean ≈ $f(x)$
no
B
yes
yes
coverage ≈ nominal at k=1 and k=2?
no
C
A
yes
Report: $y \pm U(y)$, k and coverage probability; same-instrument and transfer budgets separately; how u_{rand} and u_{sys} were obtained, stationarity of u_{sys}; full covariance for correlated outputs; linearity, MC sample size M, input distributions
regression
classification
both tracks
decision
validation return to:
A Phase 3: adopt MC as primary method (linearity violated)
B Phase 2: revise the input-uncertainty model
C Phase 2: under-coverage, missing source; over-coverage, input part overstated